

Characteristic-Time Analysis: Retained Mass versus Microtubule Count

15 August 2026 · grid dependence added 27 August 2026

Parameters

Common to every run in both tables:

Advective velocity	$v = 10^4$
Switch rates	$a = b = w = 100$
Dimensionless solution time	$T = 1$
Domain radius	$R = 1$
Diffusion coefficient	$D = 1$
Microtubule extraction width	$d_{\text{tube}} = 0$
Initial condition	centred

N is the number of evenly spaced microtubules, t^* the characteristic time,
and m^* the retained mass evaluated at t^* .

Domain grid: 96 x 96

N	t^*	m^*	Wall time (hr)
4	0.073186	0.057734	10.61
8	0.067073	0.147390	10.72
16	0.059530	0.377297	10.88
24	0.054723	0.578565	11.09

Domain grid: 48 x 48

N	t^*	m^*	Wall time (min)
4	0.070148	0.096793	10.5
8	0.061825	0.295470	10.7
16	0.052357	0.670035	11.1
24	0.046611	0.858021	11.6

Grid dependence at $N = 16$

The same parameters, with the microtubule count held at $N = 16$ and the grid varied. The 48 x 48 and 96 x 96 rows reproduce the $N = 16$ rows of the tables on the previous page. m is the slope of $\log_{10}$ of total mass across the fit window t in $[0.4, 0.5]$, from which t^* is derived.

Grid	t^*	m^*	m	Device	Wall time
48 x 48	0.052357	0.670035	-0.369056	CPU	3.8 min
64 x 64	0.055390	0.546092	-0.527038	CPU	19.3 min
80 x 80	0.057689	0.450505	-0.666989	CPU	1.23 hr
96 x 96	0.059530	0.377297	-0.790103	CPU	3.68 hr
112 x 112	0.061058	0.320804	-0.898529	CPU	9.32 hr
128 x 128	0.062355	0.276652	-0.994425	CPU	21.31 hr
144 x 144	0.063476	0.241655	-1.079686	CPU	43.68 hr
160 x 160	0.064458	0.213518	-1.155910	CPU	82.36 hr
176 x 176	0.065327	0.190594	-1.224422	GPU	22.48 hr
192 x 192	0.066104	0.171680	-1.286315	GPU	30.32 hr
208 x 208	0.066804	0.155899	-1.342492	GPU	35.26 hr
224 x 224	0.067439	0.142591	-1.393708	GPU	54.01 hr